Project Title: Swim Team Scheduling and Performance Platform for University of Evansville Swimming & Diving

Project Overview: The goal of this project is to build a web-based platform for scheduling and performance tailored for the University of Evansville Swimming & Diving team. The system will have academic, athletic, and team schedules in one place, while also automatically identifying scheduling conflicts. This will help coaches determine optimal practice times with the fewest conflicts more efficiently as this is done manually by hand currently. This application will provide swimmers and coaches with access to performance information, goals, and other useful team data in one reliable place. This project is intended to reduce the amount of information that is spread across separate systems and make scheduling and performance tracking easier and more efficient for both coaches and swimmers.

Users/Stakeholders: The primary client would be the University of Evansville Swimming & Diving coaching staff, particularly the head coach. Swimmers would also be major users and stakeholders of the system. I already have direct access to the coaching staff and swimmers, which will make communication, requirements gathering, feedback, and testing much easier. I have discussed the project and draft proposal with my coach, and he responded positively to the direction and current feature set.

User Stories:

1. As a Coach, I want to be able to view all my athletes' schedules and easily view conflicts, so that I can identify practice times with the fewest conflicts.

2. As a Swimmer, I want to be able to view my schedule, practices, meets, travel, and other team activities in one place, so that I can stay organized.

3. As a Coach, I want the system to automatically identify scheduling conflicts so that I do not have to manually compare every athlete's schedule.

Functional Requirements:

- The system must allow coaches and swimmers to log in.
- The system must support different permissions for coaches and swimmers.
- The system must store class schedules and team activities.
- The system must automatically identify scheduling conflicts.
- The system must calculate/rank practice windows with the fewest conflicts.
- The system must store swimming results and personal best times.
- The system must display swimmer performance progression.
- The system must allow users to create and track goals.
- The system should allow optional practice/training information to be recorded.

Non-Functional Requirements:

- Conflict detection should be accurate for correctly entered schedules.
- Low-conflict practice windows should be generated within 10 seconds.
- Student schedules and performance information must be protected.
- The interface should be simple enough for coaches and swimmers to use without significant training.
- The system should work on common desktop and mobile web browsers.
- The design should follow basic accessibility practices.

Client Interview Questions:

- **The Real Problem:**
  -From your perspective, how do you currently feel about how scheduling, performance information, and other team information are currently handled; what are the biggest problems currently?
- **Workflow:**

-Can you walk me through how you currently decide on practice times and find the best options for the greatest number of swimmers?
-Can you tell me about how you currently look at your swimmer's progress and what tools you currently use?
-How long does it normally take for you to recognize a scheduling conflict, and how long does it normally take to resolve?

- **Constraints:**
  -What information should a coach be able to see, that student athletes should not be able to see?
  -Who, if anyone, outside of the team should have access to this information and what should they be able to see?
- **Success Criteria:**
  -If this system were fully functional, what would success look like to you? What would you want it to make easier, faster, or better for the team?

Responsible Computing Concerns: The main concerns are privacy and security for this project. The system could potentially contain class schedules, performance data, goals, and other information about student-athletes. I would have to ensure that a coach versus an athlete have different access to certain features, and I would also have to avoid using performance statistics in ways that create unfair comparisons between athletes.

Prioritized Features:

Must Have Features:

- Unified Scheduling and Calendar
- Automatic Conflict Detection
- Low Conflict Practice-Time Dashboard
- Performance Tracking and Analytics
- Coach and Swimmer Accounts with Appropriate Permissions

Nice to Have Features:

- Athlete/Team Goal Tracking
- More Detailed Performance Graphs and Filtering
- File/CSV Imports for Existing Performance or Schedule Data

Features That May Be Postponed:

- Optional Practice Logs
- Notifications/Reminders
- Integration With Already Existing Platforms like ARMS & SwimCloud

Open Questions/Risks:

- **Scope:** There are many features that could be added to this project, but it must stay focused on the high priority requirments initially.
- **Data access/integration:** It is still unclear how easy it will be to get schedule and performance data from already existing platforms. Some manual entry or file imports may be required, but I can get everything from my client if need be.
- **Privacy & Permissions**: The exact information coaches, swimmers, and people outside the team should be able to view still needs to be confirmed with the client.

- **Client Workflow:** Further discussion with the coach is needed to fully understand the current scheduling process and determine which parts of the system would provide the most value.